Curated Research Articles

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- **A Multimodal In-Situ Liquid Phase TEM and Electrochemical Study on Boric Acid as Electrolyte Additive for Alkaline Zinc-Air Batteries** — score: 1.000 The study investigates the enhancement of alkaline zinc-air battery performance through the addition of boric acid as an electrolyte additive, using a multimodal approach that integrates nanoscale and macroscale analyses. Findings reveal that boric acid mitigates hydrogen evolution, promotes uniform zinc deposition, and significantly reduces corrosion, leading to improved cycling efficiency and battery longevity. These insights connect nanoscale phenomena to overall device performance, underscoring the potential of boric acid in practical applications. Journal: *Wiley: Small Methods: Table of Contents*
- **Unveiling Hidden Patterns in Electrode-Electrolyte Interphase Chemistry and Architecture Through Machine Learning-Assisted Mass Spectrometry Analysis** — score: 1.000 The article explores the use of machine learning to analyze 3D data from time-of-flight secondary ion mass spectrometry (TOF-SIMS) for understanding the complex architecture of electrode-electrolyte interphases in batteries. By applying advanced data analysis techniques such as principal component analysis and t-distributed stochastic neighbor embedding, the study reveals patterns in chemical composition and spatial organization that influence battery performance, ultimately contributing to the design of more uniform and stable interphases. Journal: *Wiley: Small Methods: Table of Contents*
- **Dual Descriptor-Guided Design of Hydrated Eutectic Electrolytes for Stable Zinc Batteries** — score: 1.000 This study introduces effective hydroxyl density (EOD) and minimum electrostatic potential (MEP) as quantitative descriptors to evaluate polyol hydrogen-bond donors (HBDs) in zinc-salt-based eutectic electrolytes. Among the tested polyols, erythritol demonstrated optimal performance by providing stable interfacial dynamics, leading to significant improvements in zinc battery cycling stability and suggesting a systematic approach for future electrolyte designs. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Facet-Dependent Plating/Stripping Behavior and Crystallographic Evolution during Cycling on Zn Metal Anodes** — score: 1.000 This study investigates the crystallographic influence on zinc plating and stripping behaviors in aqueous zinc metal batteries using single-crystal electrodes with different facets. The researchers found that zinc deposition and dissolution are highly dependent on the crystalline orientation, leading to distinct growth morphologies and dissolution patterns. However, repeated cycling leads to uneven zinc growth, resulting in dendritic structures and loss of efficiency, underscoring the importance of managing crystallographic factors during battery cycling. Journal: *ACS Energy Letters ASAP*
- **Surface Fluorination of LiNi 0.5 Mn 1.5 O 4 by Fluoropolymer Thermal Decomposition: From Reactive Gas Speciation to Electrochemical Stabilization** — score: 1.000 This study explores the surface fluorination of LiNi0.5Mn1.5O4 (LNMO) cathodes using two fluoropolymer agents, PTFE and CYTOP, revealing their effects on electrochemical stability and surface chemistry. Both treatments led to significant improvements in capacity retention after cycling, attributed to differences in the reactive species produced during thermal decomposition, ultimately providing insights for designing effective fluoropolymer-based surface modifications for battery materials. Journal: *ACS Applied Energy Materials ASAP*
- **Persistent Inter-particle State-of-Charge Heterogeneity in Fast-Charged High-Loading NMC811 Electrodes Revealed by Ex Situ Raman Spectroscopy** — score: 1.000 The study investigates the non-uniform lithiation and degradation in high-loading NMC811 electrodes used in lithium-ion batteries, utilizing ex situ Raman spectroscopy to analyze the state-of-charge (SOC) heterogeneity. Findings reveal that fast charging rates create a subgroup of particles with near-0% SOC, persisting even after rest, suggesting that rapid charging can exacerbate degradation and contribute to capacity loss by isolating certain particles kinetically. Journal: *ACS Applied Energy Materials ASAP*
- **Heteroatom-Mediated Functional Carbon Engineering for the Simultaneous Enhancement of Graphitization and Interfacial Stability in LiFePO 4** — score: 1.000 The study introduces

a carbon engineering method using LiBOB and LiBF4 to enhance both the graphitization of carbon coatings and the interfacial stability of LiFePO4 (LFP) cathodes. The incorporation of these precursors results in superior high-rate performance, improved cycling stability, and reduced charge-transfer resistance by strengthening the cathode/electrolyte interface, particularly with the LiBF4 modification showing the most significant benefits. This research emphasizes the dual role of functionalized carbon layers in enhancing both conductivity and interfacial chemistry. Journal: *ACS Applied Energy Materials ASAP*

- **Selective Interfacial Coupling in Polymer Electrolytes Stabilizes High-Voltage Ni-Rich Cathodes for > 500 Wh Kg−1 Lithium Metal Batteries** — score: 1.000 The article presents a novel approach to enhancing the performance of high-voltage lithium metal batteries by employing a PCA-DFA polymer electrolyte that achieves selective interfacial coupling. This design stabilizes critical cathode/electrolyte interactions, leading to a lithium metal pouch cell with an impressive energy density of 529.4 Wh kg−1 and 91.4% capacity retention after 65 cycles, showcasing its viability for advanced battery applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Anion-Driven Solvent Degradation Networks in High-Concentration Battery Electrolytes** — score: 1.000 The study investigates the reaction mechanisms behind the formation of robust solid electrolyte interphases in high-concentration battery electrolytes, utilizing automated discovery and microkinetic modeling. It identifies that specific anions, particularly reduced fluorides and trifluoromethyl groups, significantly influence solvent degradation pathways and product distributions, underscoring the importance of anion-solvent pairing in optimizing electrolyte performance. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Near-Micron Single-Crystalline Mn-Rich Cathodes With Exceptional Cycling Stability** — score: 1.000 This study highlights the development of near-micron single-crystalline Mn-rich -phase cathodes with impressive energy density of approximately 640 Wh kg−1 and 97% capacity retention over 400 cycles. The researchers used high-throughput automated electron microscopy to identify compositional heterogeneities that lead to impurities, and employed rapid cooling techniques to optimize particle size and reduce synthesis times, thus enhancing the performance of Mn-redox-based lithium-ion batteries, marking progress towards more sustainable battery materials. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Interfacial Dipole Reorganization Enables Ultrafast Sodium Storage in Na4Fe3(PO4)2P2O7 Cathodes** — score: 1.000 The study explores a novel approach to improve the sodium storage capabilities of Na4Fe3(PO4)2P2O7 by introducing Ni2+ to enhance electronic interactions and interfacial dipole reorganization. This process results in significant improvements in Na+ exchange kinetics and reduced polarization, achieving a capacity of 118 mAh g−1 and stable performance over 10,000 cycles, demonstrating the potential for practical application in sodium-ion batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Stress Regulation through Precursor Chemistry: From Synthesis to Cycling Stability in O3-Type Layered Oxide Cathodes** — score: 1.000 The article presents a novel precursor design strategy to address stress accumulation in O3-type layered oxide cathodes used in sodium-ion batteries. By utilizing low-melting-point metal salts during synthesis, the approach achieves enhanced structural stability and a remarkable 25.65% increase in cycling capacity retention, highlighting the importance of uniform residual stress distribution for improved electrochemical performance. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **High-Resolution 6 Li Solid-State Nuclear Magnetic Resonance Reveals Hidden Lithium Dynamics in All-Solid-State Cathodes** — score: 1.000 The article discusses a novel approach using high-resolution ^6Li solid-state nuclear magnetic resonance (NMR) to investigate the dynamics of lithium in iron-based halide cathodes within all-solid-state lithium batteries. It identifies distinct lithium environments and the influence of local disorder on redox behavior, revealing critical information about lithium distribution that affects battery performance, including enhancements in cycling stability and capacity retention. This technique highlights the significance of hidden lithium dynamics in optimizing cathode materials for better battery efficiency. Journal: *Journal of the American*

Chemical Society ASAP

- **Ammonium Ion Intercalation Enabled Stable 1T-VS 2 /WS 2 -Nb 2 CT x Heterostructures during Electrochemical Charge Storage** — score: 1.000 The study investigates the stabilization of 1T-VS /WS -Nb CT heterostructures through ammonium ion intercalation, enhancing their electrochemical performance. The modified electrodes achieved a specific capacity of 217.48 mA h g ^{1}, showcasing excellent stability during high-density cycling, supported by simulations and operando analysis that elucidate the interactions and dynamics of the intercalated ions. An asymmetric supercapacitor based on this configuration demonstrated a peak energy density of 54 Wh kg ^{1}, indicating promising applications in energy storage. Journal: *ACS Energy Letters ASAP*
- **Bulk Electronic Transformations in Si–Ge Alloys during Lithiation Probed by Si L 2,3 -edge X-ray Raman Scattering** — score: 1.000 This study employs Si L2,3-edge X-ray Raman scattering (XRS) to explore how silicon-germanium (Si-Ge) alloys undergo electronic transformations during their initial lithiation cycle. By analyzing both micron/submicron and nanosized Si0.5Ge0.5 electrodes, the research reveals how the presence of germanium alters the electronic structure of these materials, emphasizing the complexity of lithium interactions and the impact of alloy composition and morphology on their performance as anodes in lithium-ion batteries. Journal: *ACS Applied Energy Materials ASAP*
- **Spontaneous Formation of a Li–Al Oxide Coating Layer on Ni-Rich Positive Electrode Materials and Improvement of Electrochemical Performance** — score: 1.000 This study demonstrates a novel and eco-friendly method for enhancing the electrochemical performance of Ni-rich lithium-ion battery positive electrodes by mixing them with Al2O3 nanoparticles. The residual alkali species on the electrode surface spontaneously react with the nanoparticles to form a protective Li–Al oxide coating, which significantly improves the material's capacity retention and reduces surface degradation during cycling. Journal: *ACS Applied Energy Materials ASAP*
- **Cooperative Bulk-Interface Regulation Through Na/F/Al Functional Synergy Enables Structural and Interfacial Stability in Li-Rich Layered Oxides** — score: 1.000 The article presents a method for enhancing the performance of lithium-rich manganese layered oxides (LRMs) as cathode materials in lithium-ion batteries by employing a synergistic approach using Na+, F−, and Al3+. This combination improves lithium transport, stabilizes the chemical structure, and creates a robust cathode electrolyte interphase, leading to high discharge capacities and impressive cycling stability, thereby addressing previous challenges in the material's practical application. Journal: *Wiley: Small Methods: Table of Contents*
- **Evidence for multi-Li+ clustering in lithium-ion battery electrolytes** — score: 1.000 The article discusses groundbreaking findings on multi-lithium-ion complexes identified through machine learning-assisted atomistic simulations and various analytical methods. These complexes, which lack bound anions, enhance the understanding of the atomic behaviors influencing lithium-ion battery performance. Journal: *Nature*
- **Pressure-Driven Topological Reconfiguration toward Fast-Charging Hard Carbon with Plateau-Capacity Retention** — score: 1.000 The article presents a novel method for enhancing hard carbon performance by utilizing a pressure-driven thermal sintering technique, which facilitates the formation of interconnected pseudo-graphitic structures with nanopores. This approach not only accelerates sodium ion transport, allowing for rapid charging, but also ensures high retention of plateau capacity, significantly improving the material's efficiency for energy storage applications. Journal: *ACS Energy Letters ASAP*
- **Grain Boundary Character Governs Lithium Deposition Uniformity on Polycrystalline Copper** — score: 1.000 The study investigates how the microstructure of polycrystalline copper current collectors influences lithium deposition in anode-free lithium metal batteries. It reveals that grain boundary density and character significantly impact the uniformity of lithium nucleation and growth, with coarser grains promoting better deposition, while disordered boundaries create barriers that hinder uniform plating. Optimized copper foils enhance performance across various electrolytes,

contributing to improved cyclability. Journal: *ACS Energy Letters ASAP*

- **Pressure–Temperature Regulation of Chemo-Mechanical Degradation in Ni-Rich Composite Cathodes for All-Solid-State Batteries** — score: 1.000 The study investigates how pressure and temperature affect the mechanical degradation of Ni-rich composite cathodes in all-solid-state batteries, revealing that an optimal operational range of 30–70 MPa and 20–50 °C can enhance electrochemical performance while mitigating issues like particle cracking and interfacial debonding. Utilizing an advanced phase-field framework, the research demonstrates that while increased pressure can enhance charge capacity by preserving interface contact, excessive pressure and temperature may lead to intensified mechanical damage and diminished battery performance. Journal: *ACS Applied Energy Materials ASAP*
- **Tailored ZnO/Cellulose Separator Reconfigures Solvation Chemistry for Durable Magnesium Metal Batteries** — score: 0.900 This study introduces a novel cellulose and zinc oxide (ZnO) separator that significantly improves magnesium metal battery performance by enhancing solvation chemistry. The separator's design effectively anchors solvent molecules, leading to a more stable magnesium ion solvation sheath and the formation of a dense solid electrolyte interphase. This innovative approach results in symmetric cells with over 2000 hours of lifespan, minimal overpotential, and compatibility with various electrolytes, promoting the potential for long-lasting and safe magnesium battery applications. Journal: *Journal of the American Chemical Society ASAP*
- **Interlayer-Engineered 1T-MoSe2 Enabling Reversible Mn2+/H+ Hybrid Storage for Aqueous Manganese Batteries** — score: 0.900 The article presents an innovative approach to aqueous manganese batteries using interlayer-engineered 1T-phase MoSe2, which enhances ion intercalation and transport due to increased interlayer spacing. This modification facilitates fast, reversible storage for Mn2+/H+ ions, achieving high capacity retention after extensive cycling, thus showing promise for safer, cost-effective energy storage solutions. Journal: *Wiley: Small Methods: Table of Contents*
- **Addressing the Zinc Corrosion Conundrum: The Vital Role of Resting-State Kinetics** — score: 0.800 The article explores the significant issue of corrosion in aqueous zinc metal batteries (AZMBs), which impacts their anode performance and overall efficiency. It presents a new mechanistic framework based on electrochemical principles to understand galvanic corrosion and its relation to hydrogen evolution reactions, while suggesting future research directions to mitigate this challenge and enhance the batteries' operational stability. Journal: *ACS Energy Letters ASAP*
- **Molecular Design of Fluorinated Ester Electrolytes via Skeleton Regulation Enables Aggressive Cathode Chemistries in 5V-Class Fast-Charging Lithium-Ion Batteries** — score: 0.800 The article discusses the development of a novel trifluoromethyl-substituted carboxylate ester solvent, specifically methyl 3,3,3-trifluoropropionate (MTFP), which enhances the performance of high-voltage lithium-ion batteries using LiNi0.5Mn1.5O4 (LNMO) cathodes. By optimizing the distance between the trifluoromethyl group and the carbonyl, MTFP effectively balances lithium ion transport with oxidative stability, resulting in improved capacity retention and cycle stability during fast charging operations. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Rational Solvent Fluorination Design for Solvation Regulation Enabling Stable Interfaces in High-Voltage Li-Ion Batteries** — score: 0.800 This study explores the impact of trifluoromethyl group variations on diethyl carbonate (DEC) to improve the solvation structure of electrolytes in lithium-ion batteries under high-voltage conditions. By optimizing the number and positioning of these groups, the research identifies a specific DEC derivative that enhances interfacial stability and enables high-capacity Ni-rich cathodes to maintain performance over 300 cycles at 4.6 V. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Molecularly Aligned Insoluble Nanoaggregates for Ah-Level Aqueous Zinc-Ion Batteries** — score: 0.800 The study presents a novel molecularly aligned electrolyte (MAE) that uses insoluble ligands to create nanoaggregates in aqueous zinc-ion batteries, significantly enhancing the water-repellence of electrode interfaces. This innovative approach minimizes side reactions and dendrite

formation, resulting in impressive performance metrics, including extended cycling stability and high capacity retention over numerous charge-discharge cycles. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Self-Healing Armor Enables Electrode Integrity and Interfacial Stability for Durable Silicon Anodes** — score: 0.800 The article details the development of a self-healing polymer coating for silicon anodes that enhances electrode integrity and interfacial stability. By utilizing a fluorinated acrylate/methacrylate copolymer, the coating effectively repairs microcracks and facilitates the formation of a stable solid electrolyte interphase (SEI), leading to significant capacity retention in silicon-based batteries over extensive cycling while also improving thermal safety. Journal: *ACS Energy Letters ASAP*
- **N-Doped Carbon Nanosheet-Confined CoSe 2 Enabling High-Rate and Durable Sodium Storage** — score: 0.800 The article presents a novel CoSe2@NC nanoarchitecture designed for sodium-ion batteries, leveraging nitrogen-doped carbon nanosheets to enhance electron transport and accommodate volumetric changes. This innovative structure significantly improves the electrochemical performance of CoSe2, resulting in high reversible capacity, rate capability, and cycling stability, while also showing promise for sodium-ion capacitor applications due to its effective charge storage dynamics and fast sodium ion diffusion. Journal: *Energy & Fuels ASAP*
- **Interfacial Stability between Li-Metal/Si Composite Anode and Sulfide Solid Electrolytes in Direct Die-Pressed and Isostatically Pressed All-Solid-State Batteries** — score: 0.800 This study explores the interfacial stability between lithium-metal pre-lithiated silicon anodes and sulfide solid electrolytes in all-solid-state batteries, focusing on both direct die-pressed and isostatically pressed configurations. By optimizing conditions for lithium thickness and distribution within the silicon electrode, the researchers demonstrated enhanced cycling stability and reduced interfacial resistance, indicating a promising path for scalable, high-energy battery applications. Journal: *ACS Applied Energy Materials ASAP*
- **Multiple-Site Interaction-Driven Coordination Mechanism for Lithium-Ion Battery Cathode Recovery Using Water-Containing Intermediate-Strength Acid-Based Green Solvents** — score: 0.800 The article introduces a novel multiple-site interaction-driven coordination (MSIC) mechanism for recycling spent lithium-ion battery cathodes, utilizing water-containing intermediate-strength acid-based deep eutectic solvents. This approach significantly enhances leaching efficiencies for lithium and cobalt while mitigating the drawbacks of traditional single-site methodologies, such as low efficiency and the use of corrosive strong acids. Journal: *ACS Applied Energy Materials ASAP*
- **Engineering Intrinsic Mixed Sodium Ionic–Electronic Conductivity in a Nickel Bis(diimine) Metal–Organic Framework** — score: 0.800 This article presents the development of Ni-HITP-SO Na, a novel mixed ionic–electronic conductor (MIEC) that integrates sodium ion and electron transport within a metal-organic framework (MOF). By incorporating sulfonate side chains, the framework exhibits significant intrinsic conductivities of 0.7 S cm ^{1} for electrons and (3.7 $\pm$ 0.7) $\times$ 10 S cm ^{1} for sodium ions at room temperature, showcasing its potential for applications in electrochemical and iontronic devices without the need for additional conductive materials. Journal: *Journal of the American Chemical Society ASAP*
- **Competitive Coordination to Ether Oxygens Enables High-Ionic-Conductivity Solid Polymer Electrolytes at Room Temperature** — score: 0.750 The article discusses the enhancement of ionic conductivity in poly(ethylene oxide)-based solid polymer electrolytes through the introduction of FeF3 as a Lewis acid additive. By altering the coordination of lithium ions and promoting salt dissociation, the FeF3-PEOPVLi electrolyte achieves a notable ionic conductivity of 0.13 mS cm−1 at room temperature, leading to improved performance in solid-state lithium metal batteries with sustained capacity over extensive cycling. Journal: *Wiley: Small Methods: Table of Contents*
- **Pore-Stitching Engineering of Biomass-Derived Hard Carbon for Enhanced Sodium-Ion Storage** — score: 0.700 The article presents a novel "pore stitching" method utilizing hydroxyethyl cellulose to enhance the structural integrity and pore distribution of biomass-derived hard carbon used

in sodium-ion batteries. By facilitating the reconstruction of disrupted carbon frameworks during carbonization, this approach significantly improves sodium-storage capacity and battery performance, achieving a reversible capacity of 375.3 mAh g−1 and maintaining 97.8% capacity after 200 cycles in practical pouch cells. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Dense Lithium Deposition on Ultrathin Eutectic Sn–Bi–In Foils by Coupled Diffusion Doping and Dynamic Creep Wrinkling** — score: 0.700 The article presents a novel approach using an ultrathin eutectic Sn–Bi–In alloy foil to enhance lithium (Li) deposition in batteries, successfully preventing dendrite formation through dynamic creep wrinkling and diffusion doping. This method results in dense, uniform Li deposits with high reversibility and prolonged cycling stability, marking a significant advancement for high-energy-density lithium metal batteries. Journal: *Wiley: Small Methods: Table of Contents*
- **Lightweight CNT Fabric Current Collectors Enabling Silicon Anodes in High-Capacity Lithium-Ion Batteries** — score: 0.700 This article explores the use of lightweight carbon nanotube (CNT) fabrics as current collectors in silicon anodes for lithium-ion batteries, replacing traditional metal collectors and enabling a significant increase in energy capacity. The study highlights that the CNT-based anodes can achieve a capacity of 1000 mAh g–1 over 700 cycles, potentially enhancing cell-level energy density by approximately 25%. Additionally, it discusses how the structure influences interfacial contact and solid-electrolyte interphase formation, affecting overall battery efficiency. Journal: *ACS Applied Energy Materials ASAP*
- **A -Conjugated Polymer Anode With Railway-Like NH4+ Pathways for High-Rate Aqueous Ammonium-Ion Batteries** — score: 0.700 The article discusses the development of a -conjugated polymer anode made from poly(5,8-diaza-1,4-naphthoquinone) (PANQ) that features engineered railway-like NH4+ transport pathways to enhance the performance of aqueous ammonium-ion batteries (AAIBs). By utilizing aligned carbonyl groups within rigid polymer structures, the electrode exhibits impressive capacity, rate capability, and cycling stability, significantly improving energy storage efficiency and electrochemical kinetics for AAIB applications. Journal: *Wiley: Small Methods: Table of Contents*
- **Redefining the thermal safety boundaries of ether-based sodium batteries** — score: 0.700 The article examines the thermal safety of ether electrolytes in sodium batteries, highlighting their superior thermal stability compared to carbonates and resilience against metallic sodium. However, it points out that this advantage is compromised by oxygen-driven autoxidation. The findings suggest that anode-free ether configurations could be the optimal future direction for sodium batteries, striking an essential balance between energy density and safety. Journal: *Joule*
- **Electrochemical and Mechanical Understanding of Extruded and Solvent-Cast Solid Polymer Electrolytes** — score: 0.600 The study evaluates the electrochemical and mechanical properties of ultrathin solid polymer electrolytes (SPEs) made from PEO–LiTFSI, comparing those produced through extrusion and solvent casting methods. Key findings indicate that extrusion improves ion transport by shortening polymer chains and that Al2O3 reinforcement enhances mechanical stability and fracture toughness, making these optimized SPEs well-suited for solid-state battery applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Dynamics and Structure of Room Temperature Ionic Liquids: Ultrafast IR Experiments and MD Simulations of the Ion Cage Model** — score: 0.600 The study investigates the dynamics and structural properties of room-temperature ionic liquids (RTILs) using ultrafast infrared spectroscopy, NMR, and molecular dynamics simulations, focusing on the role of ionic and neutral probes. By analyzing different ionic liquids with a shared cation and varying anions, the research highlights the significance of the ion cage model in understanding solvation dynamics, showing that the time for ions to escape their surrounding cage influences key properties such as diffusion and viscosity. Journal: *Journal of the American Chemical Society ASAP*
- **Exploring Local Structure in Iron Oxide Nanoparticles During Chemical Vapor Synthesis by in Situ and Operando X-Ray Absorption Spectroscopy** — score: 0.600 The article discusses

the use of in situ and operando X-ray absorption spectroscopy (XAS) to investigate the formation of iron oxide nanoparticles during chemical vapor synthesis. By employing XANES and EXAFS techniques, the study reveals dynamic oxidation state changes and local structural variations, enabling insights into the thermal behaviors of the nanoparticles in a challenging gas-phase synthesis environment. Journal: *Wiley: Small Methods: Table of Contents*

- **Sodium-Ion Battery Technology: Advances and Challenges** — score: 0.500 The article discusses recent advancements in sodium-ion battery technology, highlighting their potential as a viable alternative to lithium-ion batteries. It also addresses the challenges that still need to be overcome, such as energy density, cycling stability, and commercialization hurdles, to fully realize the potential of sodium-ion batteries in energy storage applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **3D-Nanostructured ITO Enables High-Loading Prussian Blue Cathodes for Zn-Based Electrochromic Smart Windows** — score: 0.400 The study introduces a novel approach to enhance the performance of Prussian Blue electrochromic cathodes in Zn-based smart windows by utilizing 3D-nanostructured ITO electrodes. This method allows for high-loading deposition of Prussian Blue, resulting in improved charge density, rapid color changes, and superior cyclability, which are further supported by a reversible zinc electrodeposition system. This innovation significantly boosts optical contrast and energy storage capabilities in smart window applications while extending device longevity through in-situ rejuvenation. Journal: *Wiley: Small Methods: Table of Contents*
- **Batteries that sense dendrite growth** — score: 0.400 The article discusses a novel solid electrolyte capable of detecting lithium dendrite growth in batteries by transforming this growth into an electrical signal. This innovation offers a proactive approach to battery management, allowing users to receive early warnings of potential failures, thus enhancing safety and performance in battery systems. Journal: *Nature Reviews Materials*
- **Hybrid Supercapacitor Based on Functionalized Ni-MOF Composite: Achieving Superior Power Density in Asymmetric Energy Storage Device** — score: 0.400 The study presents a novel hybrid supercapacitor using a bromine-functionalized nickel-based metal-organic framework (Ni-HBTM) combined with reduced graphene oxide, achieving a remarkable specific capacitance of 1237.3 F/g and a power density of 11,250 W/kg. This asymmetric energy storage device not only maintains 94% capacitance after 10,000 cycles but also demonstrates practical application by successfully powering LEDs, indicating its potential for advanced energy storage solutions. Journal: *ACS Applied Energy Materials ASAP*
- **Light Ignites Lattice Oxygen in an Asymmetric Motif** — score: 0.300 The study investigates the enhanced activity of lattice oxygen in a Cu-doped In2O3 system with asymmetric M1-O-M2 motifs, which facilitate oxygen activation and release under mild conditions. Utilizing real-time spectroscopy, the researchers demonstrate that illumination weakens the Cu–O bond, allowing for efficient lattice oxygen extraction, leading to notable catalytic performance for H2S detection. This work highlights the significance of local coordination asymmetry in developing photodriven catalysts. Journal: *Journal of the American Chemical Society ASAP*
- **Unlocking the Longevity of Graphite-Based Batteries in Ether Electrolytes through Solvent-Co-Intercalation-Resistant Electrolyte Design** — score: 0.300 This article discusses a novel approach to enhance the lifespan of graphite-based batteries by developing a resistant electrolyte design that incorporates solvent co-intercalation in ether electrolytes. The researchers aim to address performance degradation issues, thereby improving battery efficiency and longevity. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Fast-Charging Commercial Carbon Fluoride Secondary Batteries Achieved by Covalent Bond Reconfiguration** — score: 0.300 The article discusses advancements in carbon fluoride secondary batteries that allow for rapid charging, achieved through the innovative reconfiguration of covalent bonds within the material. This research, published in Energy Storage Materials, highlights significant improvements in charging efficiency, potentially revolutionizing commercial battery appli-

cations. Journal: *ScienceDirect Publication: Energy Storage Materials*

- **Development of Na-Containing Ce–Mn Composite Redox Materials for Integrated Thermochemical Energy Storage and Air Separation Process** — score: 0.300 This study explores the use of Na-containing Ce–Mn composite oxides for an integrated thermochemical energy storage and air separation process. By optimizing coprecipitation conditions, the researchers achieved significant redox performance, identifying key material compositions that enhanced oxygen exchange capacity while noting that manganese plays a more critical role than cerium in facilitating these reactions during cycling. Journal: *ACS Applied Energy Materials ASAP*
- **Imaging of Electrochemically Programmed Single-Molecule Reactions** — score: 0.300 The article presents electrochemically programmed single-molecule reactions (PSMR), a novel method for actively controlling single-molecule electrochemistry that allows for repeated, short reaction trials within typical solution volumes. This technique employs tailored pulse waveforms to regulate reaction parameters, facilitating detailed imaging and analysis of individual reactions using electrogenerated chemiluminescence, which enhances sensitivity to local kinetics and enables super-resolution mapping of reaction activities, thereby expanding opportunities for research in various fields of molecular electrochemistry and catalysis. Journal: *Journal of the American Chemical Society ASAP*